

NBACore v8.3 Core Infrastructure Completion Work Contract

Version

v8.3.7 - v8.3.13

Document Type

AI Development Execution Specification

Target Executor

WorkBuddy / Codex / Local AI Coding Agent

1. Project Context

NBACore is an open-source basketball analytics infrastructure project.

Current completed architecture:

```
graph TD; A[Data Import] --> B[Data Asset System]; B --> C[Formula Engine]; C --> D[Node Builder]; D --> E[Visualization Workspace]; E --> F[Workspace Core];
```

Current stage:

Architecture design completed.

Goal of this task:

Complete missing infrastructure modules before entering full MVP implementation.

2. Development Principle

DO NOT

- redesign existing architecture
- replace Formula Engine
- replace Data Asset System
- create unnecessary UI
- introduce commercial account systems
- create AI analysis functions at this stage

MUST

- maintain modular architecture
 - prioritize extensibility
 - preserve user-owned data concept
 - support future multi-league expansion
-

3. Target Completion Modules

Implement:

v8.3.7 Identity & Workspace Ownership

v8.3.8 Data Provenance System

v8.3.9 Basketball Ontology Layer

v8.3.10 Competition Rules Engine

v8.3.11 Plugin Extension System

v8.3.12 Task Execution Engine

v8.3.13 Configuration System

4. Development Priority

Phase A (Critical Foundation)

Order:

1. Data Provenance System

2. Basketball Ontology Layer

3. Configuration System

Reason:

These modules determine future data compatibility.

Phase B (Platform Expansion)

4. Competition Rules Engine

5. Plugin Extension System

Phase C (Infrastructure Optimization)

6. Identity System

7. Task Execution Engine

Module Specification

v8.3.7 Identity & Workspace Ownership

Objective

Establish data ownership and workspace isolation.

NBACore is not a data owner.

Users own datasets, formulas and analysis projects.

Required Objects

User

id

name

type
created_time

Workspace

id
owner_id
name
visibility
created_time

Permission:

private
team
public

Required Functions

Create workspace

Delete workspace

Export workspace

Share workspace

Acceptance Criteria

System can:

- create independent user workspace
- isolate private data
- export project files

v8.3.8 Data Provenance System

Objective

Every data element must have traceability.

Data Metadata

Every imported value supports:

```
value
source
owner
timestamp
confidence
verification_status
```

Example:

```
{
  "value":113,
  "source":"player_upload",
  "confidence":0.95,
  "verified":true
}
```

Required Functions

Record data source

Update confidence

Track modification history

Display provenance

Acceptance Criteria

System can answer:

"Where did this data come from?"

v8.3.9 Basketball Ontology Layer

Objective

Create universal basketball terminology mapping.

Different sources:

PTS

Points

Score

得分

Map to:

Basketball Concept:

Points

Core Objects

Statistic Definition

id

name

category

definition

formula

Example:

TS%

Definition:

$PTS / (2 * (FGA + 0.44FTA))$

Required Functions

Create metric definition

Map external field

Retrieve standard concept

Acceptance Criteria

Different datasets can describe the same basketball concept.

v8.3.10 Competition Rules Engine

Objective

Represent different basketball environments.

Supported:

NBA

FIBA

NCAA

Custom League

Competition Object

competition_id

name

country

level

ruleset

Ruleset

quarter_length

shot_clock

three_point_distance

court_type

special_rules

Acceptance Criteria

System can distinguish:

NBA

and

FIBA

without changing raw statistics.

v8.3.11 Plugin Extension System

Objective

Allow community extensions.

Plugin Structure

plugins/

adapters/

metrics/

visualization/

importers/

Plugin Types

Data Adapter

Metric Plugin

Chart Plugin

League Plugin

Acceptance Criteria

A third-party developer can add:

new league

new metric

without modifying core code.

v8.3.12 Task Execution Engine

Objective

Handle long-running operations.

Examples:

- large data import
 - formula calculation
 - report generation
-

Task Object

task_id
type
status
progress
result
created_time

Status

pending
running
completed
failed

Acceptance Criteria

Large operations do not block main application.

v8.3.13 Configuration System

Objective

Centralize system configuration.

Configuration Areas

Language

Units

Database

Plugins

Visualization

Rules

Example

```
{  
  "language":"zh-CN",  
  
  "weight_unit":"kg",  
  
  "distance_unit":"meter"  
}
```

Acceptance Criteria

System behavior can change without modifying source code.

5. Recommended Database Addition

Add:

users

```
id  
  
name  
  
type
```

workspaces

```
id  
  
owner_id
```

visibility

provenance

id

data_id

source

confidence

competitions

id

name

ruleset

plugins

id

name

version

author

tasks

id

type

status

progress

6. Coding Rules

Backend

Requirements:

- modular service structure
 - clear API boundaries
 - database abstraction
-

Frontend

At this stage:

Only create necessary interfaces.

Do not build complex UI.

Documentation

Every module must include:

README

API description

Data model

Test cases

7. Testing Requirements

Each module requires:

Unit Test

Verify core functions.

Integration Test

Verify interaction with existing modules.

Regression Test

Ensure existing v8.3 functions remain working.

8. Final Acceptance

v8.3 infrastructure completion is achieved when:

1.

Existing Formula Engine works unchanged.

2.

Existing Visualization works unchanged.

3.

External datasets have provenance.

4.

Different leagues can define rules.

5.

Users can own private projects.

6.

Third-party extensions are possible.

7.

Long operations have task management.

9. Development Philosophy

NBACore is not designed to own basketball data.

It provides:

Data Owner

NBACore Infrastructure

Basketball Intelligence

The system should allow:

players,

teams,

researchers,

analysts,

and communities

to build their own basketball knowledge.

END